

NEERAJ VOHRA

Solutions Architect | Enterprise AI & GenAI | Agentic AI | Responsible AI on GCP (Vertex AI)

Greater Noida, India | H1B Visa Holder - Authorized to Work in the U.S.

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PROFESSIONAL SUMMARY

- Solutions Architect with 25+ years of experience across enterprise architecture, cloud-native platforms, distributed systems, **microservices**, **Kubernetes**, **Java/J2EE**, and modern application architecture, with the last 3+ years focused on **Enterprise AI**, **Generative AI (GenAI)**, and **Agentic AI**.
- Hands-on architect for **AI Platform Architecture**, **Multi-Agent Systems**, **Retrieval-Augmented Generation (RAG)**, **Large Language Model (LLM)** applications, **AI Governance**, and **LLMOps**, delivering secure, scalable, production-ready **enterprise AI** solutions on **Google Cloud Platform (GCP)** and **AWS**.
- Implements **Responsible AI** on GCP using **Vertex AI** and Google Cloud APIs: **Model Armor** prompt and response sanitization, **Cloud DLP (Sensitive Data Protection)** inspection and de-identification, token cost governance with **generate_content** usage metadata and **count_tokens**, and observability with **W3C Trace Context**, **OpenTelemetry**, **Cloud Logging**, and **Cloud Trace**.
- Architected **enterprise AI** platforms using **Google ADK**, **Vertex AI**, **OpenAI Agents SDK**, **LangGraph**, **LangChain**, **Model Context Protocol (MCP)**, **Agent-to-Agent (A2A)**, and **FastAPI**.
- Hands-on across the architecture lifecycle: solution design, technical strategy, prototyping, **Python** development, code reviews, cloud-native deployment, **DevSecOps**, **Infrastructure as Code (IaC)**, and production implementation.

CORE SKILLS

Responsible AI on GCP (Vertex AI): Model Armor (sanitizeUserPrompt, sanitizeModelResponse), Cloud DLP / Sensitive Data Protection (inspect, de-identify, infoTypes), Vertex AI generate_content, count_tokens, token usage metadata, Prompt Injection and Jailbreak Defense, PII Redaction, Content Safety, AI Guardrails, AI Output Validation, Knowledge Grounding, Human-in-the-Loop Controls

AI Governance & LLMOps: Responsible AI, AI Governance, Policy-Driven AI, AI Evaluation, LLMOps, Explainable AI, Zero Trust AI, Audit Logging, Data Privacy, Risk Controls

AI Observability & Cost Engineering: W3C Trace Context (traceparent), W3C Baggage, OpenTelemetry, Google Cloud Logging, Google Cloud Trace, Cloud Monitoring, Trace-Log Correlation, Structured Logging, Token Usage Tracking, Model Latency and Error Monitoring, Prompt and Context Optimization, Intelligent Model Routing, AI Cost Optimization, Prometheus, Grafana, AWS CloudWatch

Enterprise AI & Agentic AI: Enterprise AI, Generative AI (GenAI), Agentic AI, Multi-Agent Systems, AI Platform Engineering, LLM Applications, AI Orchestration, Agent Memory, Retrieval-Augmented Generation (RAG), Human-in-the-Loop (HITL)

GenAI Frameworks & Protocols: Google ADK, Vertex AI, Gemini API, OpenAI Agents SDK, LangGraph, LangChain, MCP, A2A, FastAPI, Prompt Engineering, Prompt Management, Context Management, AI Input Pipelines, Evaluation Frameworks

Cloud & Platform Engineering: Google Cloud Platform (GCP), AWS, Kubernetes, GKE, EKS, ECS, AWS Lambda, Docker, Helm, Istio, Terraform, Infrastructure as Code (IaC)

Software & Architecture: Solution Architecture, Enterprise Architecture, Cloud Architecture, Platform Engineering, Microservices, Event-Driven Architecture, Hexagonal Architecture, Clean Architecture, SOA

Programming & APIs: Python, FastAPI, Java, Spring Boot, Spring, Hibernate, REST APIs, SOAP, XML

DevSecOps & Delivery: DevSecOps, GitOps, GitHub Actions, GitLab CI/CD, Jenkins, Ansible, Agile, Scrum, SAAFe, ServiceNow, Jira, Confluence

KEY ACHIEVEMENTS

- Reduced MTTR by 40% through AI-powered Incident Management, autonomous Root Cause Analysis, **AI Observability**, and workflow automation.
- Reduced cloud and AI operational costs by 35% through **context management**, prompt optimization, token usage tracking with **generate_content** usage metadata, **intelligent model routing**, **Terraform** automation, and **AWS ECS** optimization.
- Improved developer productivity by 50-60% through reusable AI Platform Services, **FastAPI** accelerators, **prompt management**, and AI-assisted development workflows.
- Improved release velocity by 45% through **DevSecOps**, **Infrastructure as Code**, **CI/CD**, **OpenTelemetry**, **CloudWatch**, **Prometheus**, and **Grafana**.

PROFESSIONAL EXPERIENCE

Solutions Architect

HCL Technologies Ltd. | Noida, India | McLean, USA | Jan 2022 - Present

- Architected an **Enterprise AI Platform** on **Google Vertex AI** using **Policy-Driven Hexagonal Architecture**, establishing reusable platform capabilities for secure, scalable, and governed **enterprise AI** adoption.
- Designed reusable AI Platform Services for **AI Governance, Policy Engine, Prompt Management, Context Management, Model Armor, Google Cloud DLP, AI Guardrails, AI Evaluation**, output validation, and **AI Cost Optimization**, improving developer productivity by 50-60% together with **FastAPI** accelerators.
- **Responsible AI - Model Armor**: Integrated **Model Armor APIs** (**sanitizeUserPrompt, sanitizeModelResponse**) into the **Vertex AI** request path to screen prompts and responses for prompt injection, jailbreaks, harmful content, and sensitive data, using policy-driven templates.
- **Responsible AI - Data Privacy**: Integrated **Google Cloud DLP (Sensitive Data Protection)** inspect and de-identify APIs with configurable infoTypes to detect and mask PII in user inputs, **RAG** context, and model outputs.
- **Responsible AI - Controls**: Implemented prompt and context controls, **content safety** validation, **knowledge grounding**, and **human-in-the-loop** review patterns across **Vertex AI** workloads.
- **Token Cost Governance**: Captured per-request token consumption from **Vertex AI generate_content** usage metadata and used the **count_tokens** API for pre-flight estimation, enabling budget controls, prompt and context optimization, and **intelligent model routing** that contributed to a 35% reduction in cloud and AI operational costs.
- **AI Observability & Tracing**: Implemented end-to-end observability with **W3C Trace Context (traceparent)** and **W3C Baggage** propagated across agents, **MCP** and **A2A** tool calls, and **Vertex AI** requests, instrumented with **OpenTelemetry** and exported to **Google Cloud Trace**, with structured **Cloud Logging** correlated by trace ID, plus **Cloud Monitoring, Prometheus**, and **Grafana** to track guardrail verdicts, errors, latency, and token cost.
- Architected and delivered **Agentic AI** and **Multi-Agent** solutions using **Google ADK, OpenAI Agents SDK, LangGraph, MCP, A2A, RAG, FastAPI**, and **Agent Memory** for Incident Management, Root Cause Analysis, **ServiceNow** automation, and enterprise workflow orchestration, reducing MTTR by 40%.
- Designed cloud-native AI deployment architecture across **Vertex AI, Kubernetes, GKE, AWS ECS, Docker, Helm, Istio, GitLab CI/CD**, and **OpenTelemetry**, improving scalability, resilience, deployment consistency, and operational visibility.
- Established **DevSecOps** and **Infrastructure as Code** using **Terraform** and **GitLab CI/CD**, contributing to a 45% improvement in release velocity.

Senior Technical Architect

Coforge Ltd. (formerly NIIT Technologies Ltd.) | Greater Noida, India | Jul 2008 - Jan 2022

- Architected enterprise cloud-native solutions using **Microservices, REST APIs, Kubernetes**, and **Event-Driven Architecture**, improving scalability, resilience, availability, and maintainability.
- Led cloud transformation and application modernization on **AWS** using **Amazon EKS, Amazon ECS, AWS Lambda, Terraform**, and **Infrastructure as Code (IaC)**, contributing to a 35% reduction in infrastructure costs.
- Designed and implemented **Kubernetes** platform architecture using **Helm, GitOps, Auto Scaling**, High Availability, Rolling, **Blue-Green**, and **Canary** deployment strategies, improving reliability and deployment consistency.
- Established enterprise **DevSecOps** practices using **Jenkins, GitLab CI/CD, Docker, Terraform**, and **Ansible**, reducing deployment time from days to under four hours.
- Provided technical leadership across solution architecture, cloud modernization, application engineering, deployment automation, platform engineering, and enterprise transformation.

System Analyst (SOA)

Birlasoft Ltd. | Noida, India | Jan 2007 - Jul 2008

- Led design and implementation of **Service-Oriented Architecture (SOA)** solutions using **SOAP, REST, WSDL, XML**, and **Web Services**, defining service contracts, integration patterns, and reusable APIs that improved enterprise interoperability and scalability.

IT Analyst - Java/J2EE

Tata Consultancy Services (TCS) | Lucknow, India | May 2006 - Jan 2007

- Developed enterprise applications using **Java/J2EE, REST APIs**, and Object-Oriented Design, integrating legacy systems with modern applications in cross-functional Agile teams.

Senior Software Engineer - Java

HCL Technologies Ltd. | Noida, India | Oct 2004 - Apr 2006

- Designed and developed enterprise applications using **Java**, **Spring**, **Hibernate**, and **REST APIs** across the full Software Development Life Cycle (SDLC), working with product owners, QA, and engineering teams.

Software Engineer - Java

CMC Ltd. | New Delhi, India | Dec 2000 - Sep 2004

- Developed and maintained enterprise and government applications using **Java/J2EE** and Object-Oriented Programming, contributing to production support, performance optimization, and application stability.

EDUCATION

Master of Computer Application (M.C.A.)

Maulana Azad National Institute of Technology (NIT), Bhopal, India | Jul 1997 - Jun 2000

CERTIFICATIONS

Enterprise AI & Generative AI: AWS Certified AI Practitioner | AWS Generative AI with Amazon Bedrock | Prompt Engineering with Claude AI | Agent-to-Agent (A2A) Protocol

Cloud & Solution Architecture: AWS Certified Solutions Architect - Associate | AWS Certified Cloud Practitioner | AWS Migration

Kubernetes & DevOps: Certified Kubernetes Administrator (CKA) | Certified Kubernetes Application Developer (CKAD) | DevOps Engineer

Agile & Delivery: SAFe Practitioner (SP) | Certified Scrum Master (CSM)